```
clc; clear;

% Line Data: [FromBus  ToBus  R(pu)  X(pu)]
lineData = [
    1  2  0.01  0.15;
    1  3  0.02  0.25;
    1  4  0.03  0.35;
    2  3  0.03  0.35;
    3  4  0.01  0.15;
    4  5  0.04  0.50;
];

% Number of buses
n = 5;
nline = size(lineData, 1);
Ybus = zeros(n,n);

for k = 1: nline
    from = lineData(k,1);
    to = lineData(k,2);
    R = lineData(k,3);
    X = lineData(k,4);
    Z = R + 1i*X;
    Y = 1/Z;
    Ybus(from, to) = Ybus(from, to) - Y;
    Ybus(to, from) = Ybus(to, from) - Y;
end
    disp(Ybus);
```

```
  Columns 1 through 4

   0.0000 + 0.0000i  -0.4425 + 6.6372i  -0.3180 + 3.9746i  -0.2431 + 2.8363i
  -0.4425 + 6.6372i   0.0000 + 0.0000i  -0.2431 + 2.8363i   0.0000 + 0.0000i
  -0.3180 + 3.9746i  -0.2431 + 2.8363i   0.0000 + 0.0000i  -0.4425 + 6.6372i
  -0.2431 + 2.8363i   0.0000 + 0.0000i  -0.4425 + 6.6372i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i  -0.1590 + 1.9873i

  Column 5

   0.0000 + 0.0000i
   0.0000 + 0.0000i
   0.0000 + 0.0000i
  -0.1590 + 1.9873i
   0.0000 + 0.0000i
```

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